

Class08: Breast Cancer Analysis Project

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class. We will extend what we learned by combining PCA as a preprocessing step to clustering using data that consist of measurements of cell nuclei of human breast masses.

The data itself comes from the Wisconsin Breast Cancer Diagnostic Data Set first reported by K. P. Benne and O. L. Mangasarian: “Robust Linear Programming Discrimination of Two Linearly Inseparable Sets”.

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Data import

The data is available as a CSV from the class website:

```
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"

# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)

head(wisc.df)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
84358402	M	20.29	14.34	135.10	1297.0
843786	M	12.45	15.70	82.57	477.1

	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean
842302	0.11840	0.27760	0.3001	0.14710
842517	0.08474	0.07864	0.0869	0.07017
84300903	0.10960	0.15990	0.1974	0.12790
84348301	0.14250	0.28390	0.2414	0.10520
84358402	0.10030	0.13280	0.1980	0.10430
843786	0.12780	0.17000	0.1578	0.08089

	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419		0.07871	1.0950	0.9053
842517	0.1812		0.05667	0.5435	0.7339
84300903	0.2069		0.05999	0.7456	0.7869
84348301	0.2597		0.09744	0.4956	1.1560
84358402	0.1809		0.05883	0.7572	0.7813
843786	0.2087		0.07613	0.3345	0.8902

	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
84358402	94.44	0.011490	0.02461	0.05688	0.01885
843786	27.19	0.007510	0.03345	0.03672	0.01137

	symmetry_se	fractal_dimension_se	radius_worst	texture_worst
842302	0.03003	0.006193	25.38	17.33
842517	0.01389	0.003532	24.99	23.41
84300903	0.02250	0.004571	23.57	25.53
84348301	0.05963	0.009208	14.91	26.50
84358402	0.01756	0.005115	22.54	16.67

843786	0.02165	0.005082	15.47	23.75
	perimeter_worst	area_worst	smoothness_worst	compactness_worst
842302	184.60	2019.0	0.1622	0.6656
842517	158.80	1956.0	0.1238	0.1866
84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663
84358402	152.20	1575.0	0.1374	0.2050
843786	103.40	741.6	0.1791	0.5249
	concavity_worst	concave.points_worst	symmetry_worst	
842302	0.7119	0.2654	0.4601	
842517	0.2416	0.1860	0.2750	
84300903	0.4504	0.2430	0.3613	
84348301	0.6869	0.2575	0.6638	
84358402	0.4000	0.1625	0.2364	
843786	0.5355	0.1741	0.3985	
	fractal_dimension_worst			
842302	0.11890			
842517	0.08902			
84300903	0.08758			
84348301	0.17300			
84358402	0.07678			
843786	0.12440			

Make sure we do not include sample id or diagnosis columns in the data that we analyze below.

```
diagnosis <- as.factor(wisc.df$diagnosis)
wisc.data <- wisc.df[,-1]
dim(wisc.data)
```

```
[1] 569 30
```

Exploratory data analysis

Q1. How many observations are in this dataset?

```
dim(wisc.data)
```

```
[1] 569 30
```

There are 569 observations in this data set.

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
  B    M  
357 212
```

There are 212 observations that have a malignant diagnosis.

Q3. How many variables/features in the data are suffixed with `_mean`?

```
#colnames(wisc.data)  
  
length(grep("_mean", colnames(wisc.data)))
```

```
[1] 10
```

There are 10 variables/features in the data set that are suffixed “`_mean`”.

Principal Component Analysis

The main function in base R for PCA is called `prcomp()`. An optional argument `scale.` should nearly always be switched to ‘`scale. = T`’) for this function.

```
wisc.pr <- prcomp(wisc.data, scale = T)  
summary(wisc.pr)
```

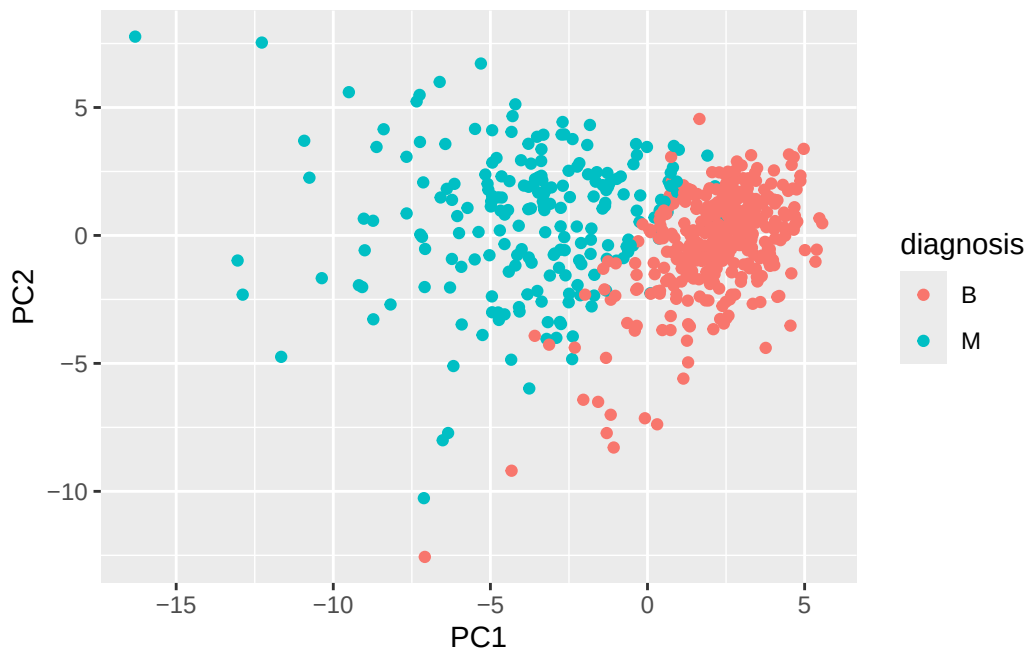
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010

Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Let's make our main result figure - the "PC plot" or "score plot", "orriention plot"

```
library(ggplot2)
ggplot(wisc.pr$x) +aes(PC1, PC2, col = diagnosis) +geom_point()
```



Q4. From your results, what proportion of the original variance is captured by the first principal components (PC1)?

```
wisc_pr_summary <- summary(wisc.pr)
wisc_pr_summary
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

The first principal captures 0.4427 of the variance.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

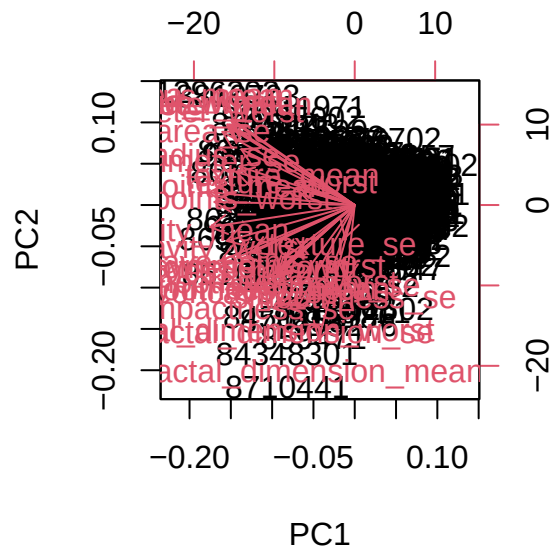
Based on the cumulative proportions of each principal components, we need the first 3 principal components to describe at least 70% of the original variance in the data since PC3 is 0.72636.

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

Seven principal components are necessary to describe at least 90% of the original variance in the data. The cumulative proportion of PC7 is 0.91010.

Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

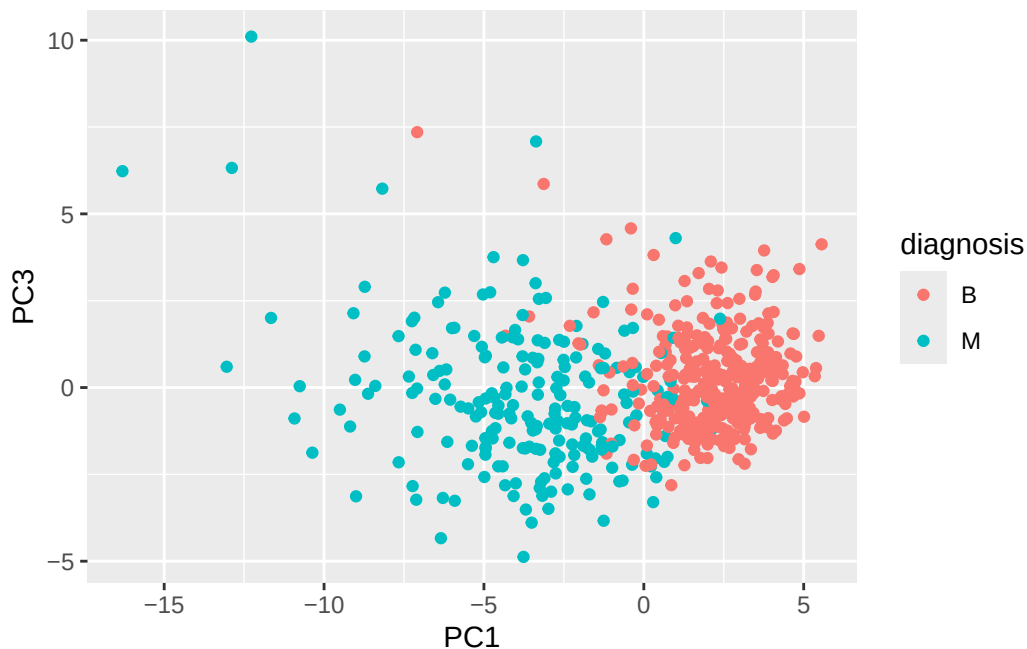
```
biplot(wisc.pr)
```



It is very difficult to read and understand this plot so we can not make any conclusions from it. This is a result of the data set being very large.

Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x) +aes(PC1, PC3, col = diagnosis) +geom_point()
```



Changing the comparison to PC3, brings the points down closer to $y = 0$. The spread of PC2 is broader (range of y axis 5 to -10) compared to PC3 (range of y axis is 5 to -5). The overall clusters in the PC1 vs PC3 also are less obvious than those in PC1 vs PC2.

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC.

```
wisc.pr$rotation[,1]
```

radius_mean	texture_mean	perimeter_mean
-0.21890244	-0.10372458	-0.22753729
area_mean	smoothness_mean	compactness_mean
-0.22099499	-0.14258969	-0.23928535
concavity_mean	concave.points_mean	symmetry_mean
-0.25840048	-0.26085376	-0.13816696
fractal_dimension_mean	radius_se	texture_se
-0.06436335	-0.20597878	-0.01742803
perimeter_se	area_se	smoothness_se
-0.21132592	-0.20286964	-0.01453145
compactness_se	concavity_se	concave.points_se
-0.17039345	-0.15358979	-0.18341740
symmetry_se	fractal_dimension_se	radius_worst

-0.04249842	-0.10256832	-0.22799663
texture_worst	perimeter_worst	area_worst
-0.10446933	-0.23663968	-0.22487053
smoothness_worst	compactness_worst	concavity_worst
-0.12795256	-0.21009588	-0.22876753
concave.points_worst	symmetry_worst	fractal_dimension_worst
-0.25088597	-0.12290456	-0.13178394

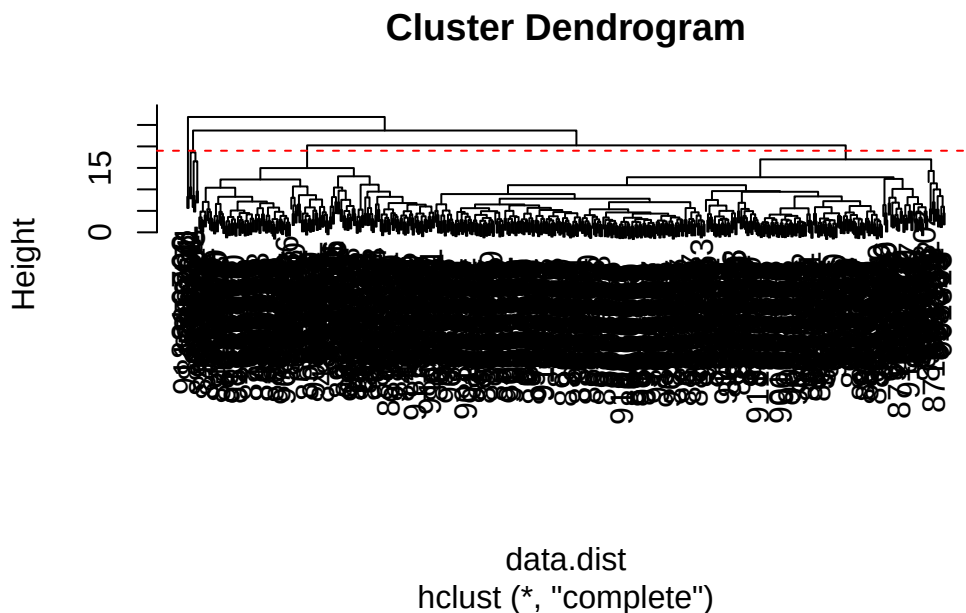
The component of the loading vector of concave.points_mean is -0.26085376 in PC1.

Heirarchial Clustering

```
# Scale the wisc.data data using the "scale()" function
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
wisc.hclust <- hclust(data.dist, method = "complete")
```

Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

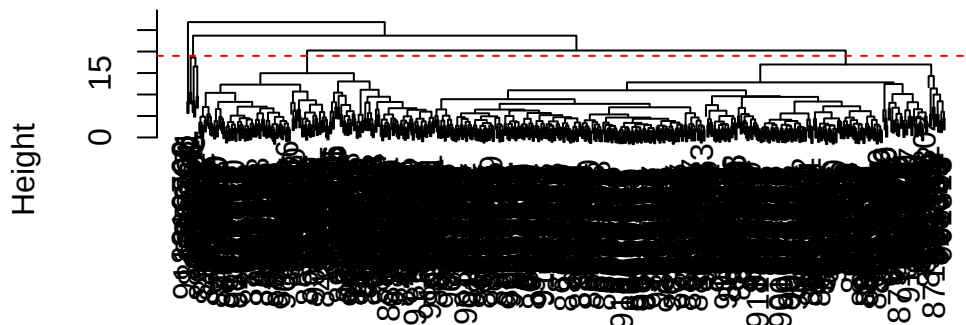
```
plot(wisc.hclust)
abline(h = 19, col="red", lty=2)
```



The height at which the clustering model has 4 clusters is 19.

```
plot(wisc.hclust)
abline(h = 19, col="red", lty=2)
```

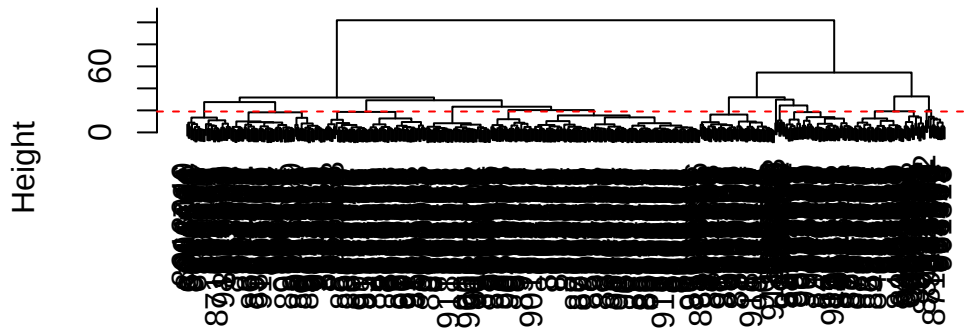
Cluster Dendrogram



```
data.dist
hclust (*, "complete")
```

```
wisc.hclust <- hclust(data.dist, method = "ward.D2")
plot(wisc.hclust)
abline(h = 19, col="red", lty=2)
```

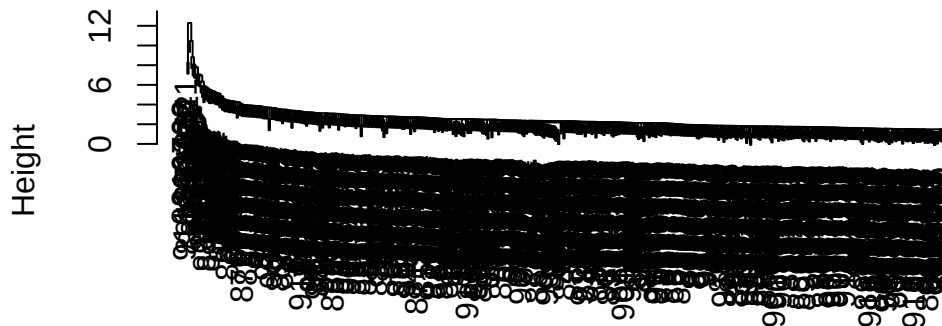
Cluster Dendrogram



```
data.dist
hclust (*, "ward.D2")
```

```
wisc.hclust <- hclust(data.dist, method = "single")
plot(wisc.hclust)
abline(h = 19, col="red", lty=2)
```

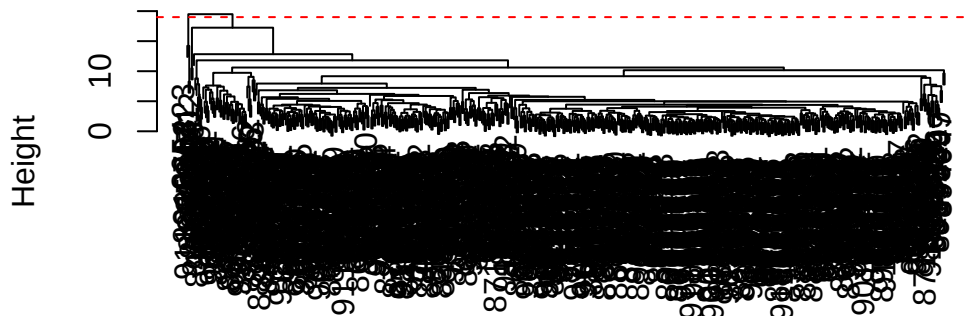
Cluster Dendrogram



```
data.dist
hclust (*, "single")
```

```
wisc.hclust <- hclust(data.dist, method = "average")
plot(wisc.hclust)
abline(h = 19, col="red", lty=2)
```

Cluster Dendrogram



data.dist
hclust (*, "average")

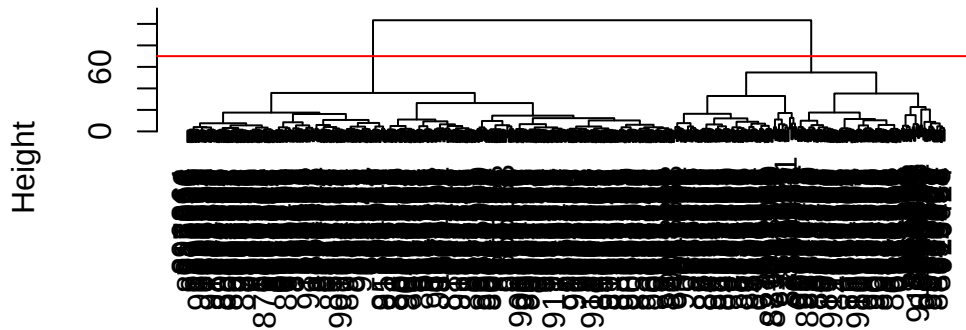
Q12. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

ward.D2 gives the best results for the same data set since it breaks the data into 2 distinct clusters. All the other methods result in dendrograms with really difficult to distinguish the different bigger clusters.

Combining PCA and Clustering

```
d <- dist(wisc.pr$x[,1:3])
wisc.pr_hclust <- hclust(d, method = "ward.D2")
#
plot(wisc.pr_hclust)
abline(h = 70, col = "red")
```

Cluster Dendrogram



d
hclust (*, "ward.D2")

Combining techniques

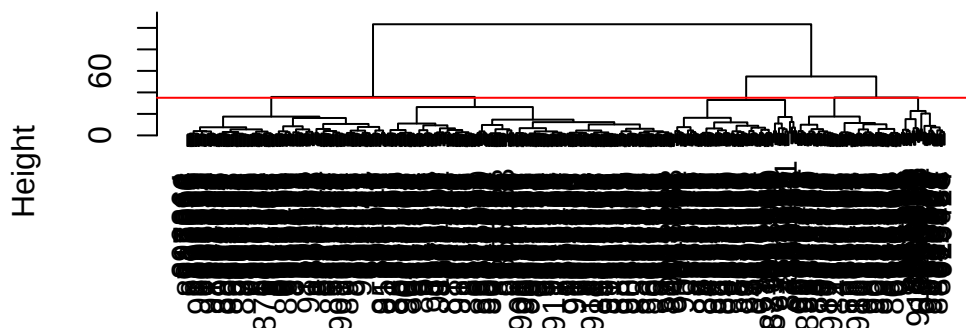
Get my cluster membership vector

```
grps <- cutree(wisc.pr_hclust, k = 2 )
table(grps)
```

```
grps
  1  2
203 366
```

```
plot(wisc.pr_hclust)
abline(h = 35, col = "red")
```

Cluster Dendrogram



d
hclust (*, "ward.D2")

Q13. How well does the newly created model with four clusters separate out the two diagnoses?

Make a wee “cross-table”

```
table(grps, diagnosis)
```

```
diagnosis
grps    B    M
1      24 179
2     333  33
```

The clustering results in 179 true positives, 24 false positives, 333 true negatives and 33 false negatives.

Sensitivity: $TP/(TP+FN) = 179/(179+33) = 0.8443$ Specificity: $TN/(TN + FN) = 333/(333+33) = 0.9098$

```
grps <- cutree(wisc.pr_hclust, k =4 )
table(grps)
```

```
grps
  1   2   3   4
111  92 216 150
```

```
table(grps, diagnosis)
```

```
      diagnosis
grps   B    M
  1     0 111
  2    24  68
  3   184  32
  4   149   1
```

The newly created model with four clusters, doesn't separate the two diagnoses super well. The first and fourth clusters separate the diagnosis pretty well with cluster one being 100% true positives malignant patients and cluster 4 being 149 out of 150 being true negative. Cluster 2 and 3 has a significant mix of both diagnoses without good separation.

Q14. How well do the hierarchical clustering models you created in previous sections (i.e. before PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.km$cluster` and `wisc.hclust.clusters`) with the vector containing the actual diagnoses.

```
hclust.single <- hclust(data.dist, method = "single")
grps <- cutree(hclust.single, k = 4 )
table(grps)
```

```
grps
  1   2   3   4
565  1   2   1
```

```
table(grps, diagnosis)
```

```
      diagnosis
grps   B    M
  1  356 209
  2    1   0
  3    0   2
  4    0   1
```

```
hclust.average <- hclust(data.dist, method = "average")
grps <- cutree(hclust.average, k =4 )
table(grps)
```

```
grps
  1  2  3  4
564  2  1  2
```

```
table(grps, diagnosis)
```

```
      diagnosis
grps   B    M
  1 355 209
  2   2   0
  3   0   1
  4   0   2
```

```
hclust.complete <- hclust(data.dist, method = "complete")
grps <- cutree(hclust.complete, k =4 )
table(grps)
```

```
grps
  1  2  3  4
177  7 383  2
```

```
table(grps, diagnosis)
```

```
      diagnosis
grps   B    M
  1  12 165
  2   2   5
  3 343  40
  4   0   2
```

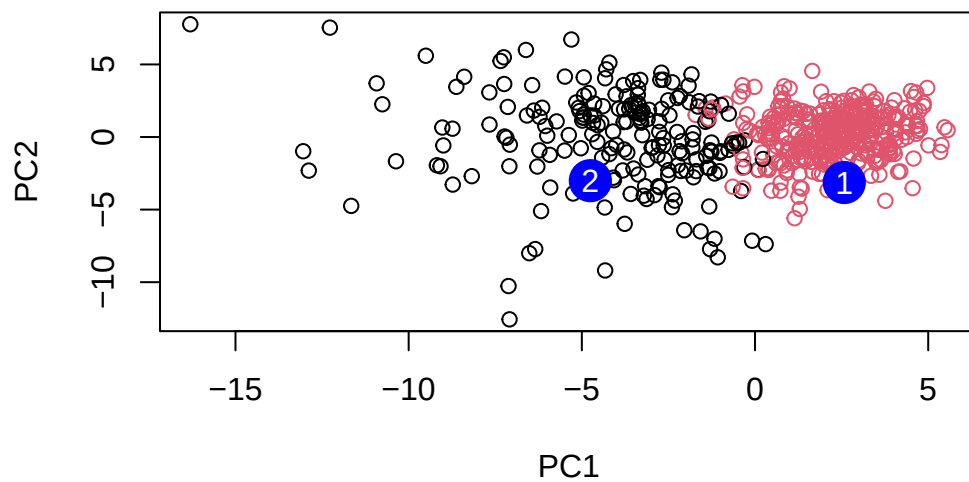
Single and average hclust hierarchical clustering techniques are definitely not better than the ward.D2 clustering method because most of the patients are clustered into one (the first) cluster. The clustering method of complete seems slightly better than single and average, however the two main clusters (1 and 3) include a mixture of patients with benign and malignant tumors in them. There are not as many patients included in cluster 2 and 4, meaning not all 4 clusters are super relevant.

Q16. Which of these new patients should we prioritize for follow up based on your results?

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
grps <- cutree(wisc.pr_hclust, k = 2)
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Based on the result, we should prioritize patient 2 for a follow up because they fall within the malignant cluster.